CLASSIFICATION CASE

[ANALYSIS REPORT]

TO:

PROF. BIN LI

FROM:

SUSHANT BHATTA - U01105863

UNISH BARDEWA - U01111591

Classification Case Analysis

Data Splitting and Optimize Parameters

The dataset was divided into training (70%) and testing (30%) samples using stratified sampling to maintain the same purchase rate across both sets, as the target variable ("Purchase") is relatively rare. In this model, "Purchase" is the dependent variable and M, R, F, FirstPurch, and Related Purchase are the independent variables utilized for prediction.

With just the training data, four distinct categorization models were constructed and refined. The parameters to be optimized are:

Decision tree: criterion, apply prepruning, **(Information Gain, False)**

Naïve Bayesian: Laplace correction, **(False)**

Neural network: learning rate, **(0.400)**

K-NN: k, **(k 15)**

2. **ROC Curve Comparison**

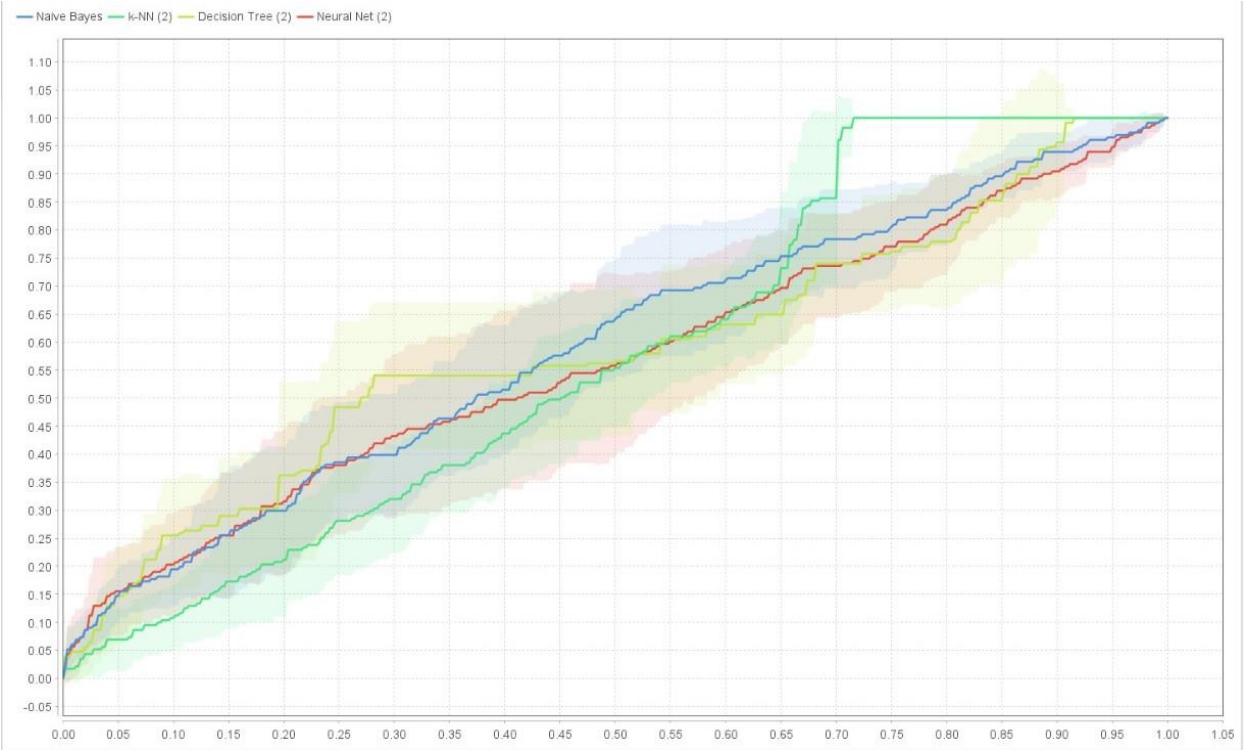

Figure 1. ROC Curve Comparison

Four classifiers—Neural Network, Decision Tree, K-Nearest Neighbors (K-NN), and Naïve Bayes—were trained and evaluated on the dataset before the ROC curve comparison was created in RapidMiner. In the ROC graph, each model is represented by a distinct color: blue is the Naïve Bayes model, light green is the Decision Tree, dark green is the K-NN model, and red is the Neural Network. Neural Network-learning rate of 0.400, Decision Tree-Information Gain criteria with prepruning set to false, K-NN model-k = 15, and Naïve Bayes model without Laplace correction were the optimized parameters of each classifier.

To determine which of these models exhibits superior prediction ability, the AUC under the ROC curve was analyzed. The K-NN model produced an AUC of 0.556, the Decision Tree recorded the lowest value of 0.529, and the Neural Network model had the greatest AUC of 0.597, followed by the Naïve Bayes model at 0.592. Among these models, a higher AUC value indicates a superior performance in terms of accurately differentiating buyers from non-buyers. Out of the four classifiers, the Neural Network demonstrated the highest overall predictive power.

The trade-off between sensitivity (True Positive Rate) and false positivity rate (1-specificity) is displayed by the ROC curve. The Neural Network's curve is found to be greater than that of all the other models, indicating that it retains a higher sensitivity while producing the same number of false positives. The Naïve Bayes model's curve is slightly lower than the Neural Network curve, despite the fact that it seems somewhat promising. Because of their poor class separation, the K-NN and Decision Tree models have less discriminative power. These findings led to the selection of the Neural Network model for further research since it achieves a superior balance between accuracy and generalization, with an AUC of 0.597.

Additionally, it has the best ability to differentiate between purchasers and non-purchasers of "The Art History of Florence." The most steady and reliable improvement over the random baseline is shown by its ROC curve. It is therefore the most appropriate and trustworthy model for additional predictive analysis and implementation.

3. Cumulative Gain Chart

According to the chart, 23 + 12 + 7 = 42 buyers from the entire sample of 107 fall into the first three deciles, or the top 50% of potential customers. In contrast to random targeting, which will only reach around 25–30% of consumers at the same level, this company may expect to reach roughly 40% of the entire number of buyers if it delivers offers to the top 50% of potential customers. The company may anticipate selling roughly 40–45% of the total books if offers are made to 50% of the prospects. This indicates that the predictive model performs a decent job of identifying the high-probability purchasers early in the campaign.

4. Lift Chart for purchases

According to the Excel data, the top 30% of customers (the first four rows) provide a Lift value of roughly 1.17 and a Cumulative Profit of 121.

According to this lift value, the organization can get a 17% greater performance than through random selection by concentrating on the top 30% of the ranked consumers. In order to increase efficiency and minimize wasted marketing effort, the model has successfully evaluated consumers such that the highest-potential purchasers are addressed first. By concentrating on the top 30% of potential consumers, the company achieves a Lift of 1.17, which indicates that the marketing campaign is outperforming random targeting by almost 17%. This demonstrates how the model may significantly raise conversion rates and marketing productivity.

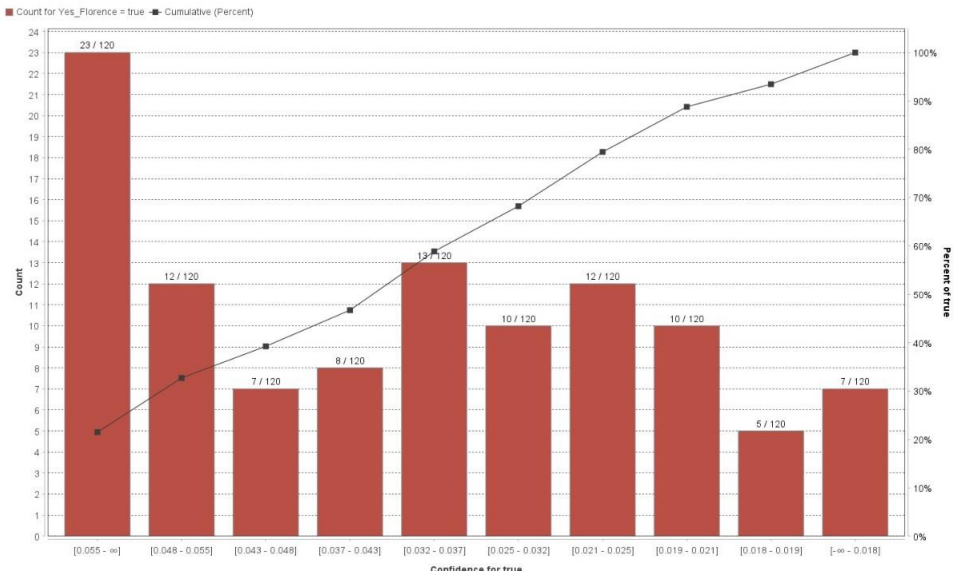

Figure 2.Lift Chart

5. Cumulative Profit Chart and Explanation

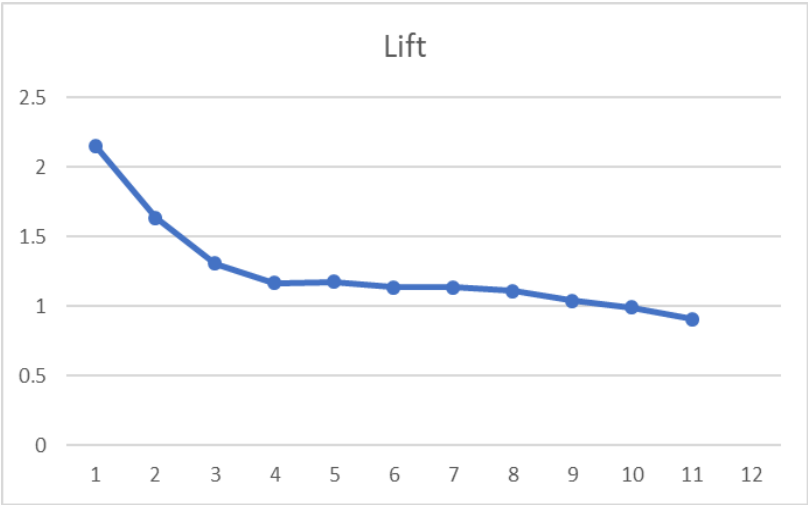

Figure 3. Chart

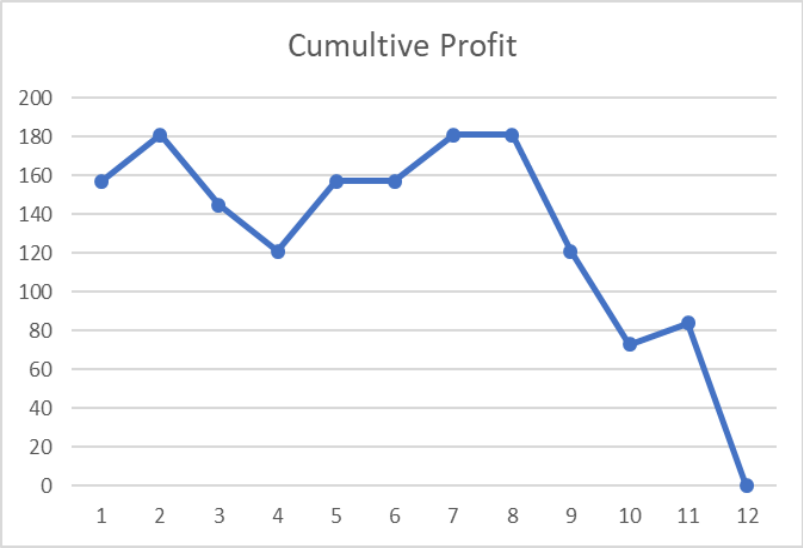

Figure 4.Cumulative Profit Chart

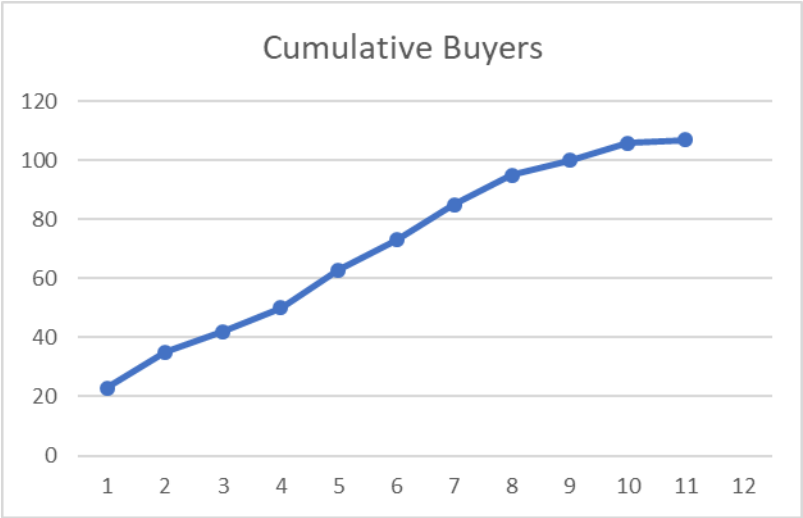

Figure 5.Cumulative Profit Chart

	A	B	C	D	E	F	G	H
1	Row Labels	Sum of profit	Sum of buyer	Cumulative Buyers	Lift	Cumultive Profit		
2	1	157	23	23	2.1495327	157		
3	2	24	12	35	1.635514	181	2nd	
4	3	-36	7	42	1.3084112	**145**	1st	
5	4	-24	8	50	1.1682243	121		
6	5	36	13	63	1.1775701	157	22625	
7	6	0	10	73	1.1370717	157		
8	7	24	12	85	1.1348465	181		
9	8	0	10	95	1.1098131	181		
10	9	-60	5	100	1.0384216	121		
11	10	-48	6	106	0.9906542	73		
12	11	11	1	107	0.9090909	84		
13								
14								
15								

Figure 6.Excel File

- **Create and report a cumulative profit chart. Explain how much the profit would be by focusing on the top 30% potential customers**.

  In this study, each customer contact costs $1.00, whereas each successful transaction generates $12.00 in revenue. According to the Excel statistics, the 4,000-customer sample's top 30% of consumers have a cumulative profit of $181. This indicates that of all the examined sectors, concentrating marketing efforts on the top 30% of consumers results in the most profitability.

- **To maximize the profit, what is the percentage of potential customers should the company send offers to? Explain your reason.**

  The first four groupings, which represent approximately 50% of potential consumers, make the highest cumulative profit of $181, according to the Excel output's cumulative profit chart. After this, the cumulative profit starts to decrease, indicating that further marketing expenses exceed the money made from focusing on lower-potential customers.

  As a result, the business should concentrate its marketing efforts on the top 50% of consumers, as this group will comprise those who are most likely to buy "The Art History of Florence." By catering exclusively to this demographic, the business may maximize profits, cut down on unnecessary marketing expenditures, and improve total ROI through effective budget allocation.

- **Given that there are 500,000 active customers in the database, how much profit can the publisher receive from this customer base based on your model?**

  The scaling method used to determine the scaled profit:

  Total Profit=181/4000×500,000=22,625

  Thus, when applying the model to all 500,000 active customers, the company can expect an **estimated total profit of $22,625**.